ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Lab 06: Comparing and Evaluating Models

Today you will use your Lab 05 results to fit a GLM and a GAM, compare their environmental response curves, and evaluate both models with the same held-out data. The goal is to ask whether a more flexible model improves prediction and to explain what we can learn from the evaluation.

Learning goals

- Separate training data from testing data before fitting models.
- Compare a linear GLM with a GAM that allows curved responses.
- Calculate held-out AUC and interpret it for presence–background data.
- Identify one useful result and one limitation for your preliminary analyses.

What you need

- Your R Project, private GitHub repository, and output/05_model_data.csv from Lab 05.
- The R package mgcv. If needed, install.packages("mgcv") once in the Console.
- Two continuous predictors with biological justification. Use the same rows, predictors, M, and background for both models.

Part A Prepare your data

1. Create analysis/06_comparing_models.R. Type and adapt the code blocks below; run from the project root.

2. Read the saved Lab 05 table. Replace bio1_c and bio12_mm with your own predictor names throughout this lab. Keep presence coded 1 for occurrences and 0 for background.

```
library(mgcv)
set.seed(689)
d <- read.csv("output/05_model_data.csv")
vars <- c("bio1_c", "bio12_mm")
stopifnot(all(d$presence %in% c(0, 1)))
stopifnot(all(is.finite(as.matrix(d[, vars]))))
table(d$presence)
```

If checks fail, return to your Lab 05 audit and diagnose the problem. Do not silently remove missing values. Background zeros are not confirmed absences.

Part B Hold out data and fit two models

Split before fitting. The code holds out approximately 30 percent of unique locations and keeps all rows at the same coordinates together. Use x and y in the same documented CRS from Lab 05; rename your coordinate columns if necessary.

```
location <- paste(d$x, d$y, sep = "_")
stopifnot(all(c("x", "y") %in% names(d)))
stopifnot(all(is.finite(d$x)), all(is.finite(d$y)))

sites <- unique(location)
train_sites <- sample(sites, floor(0.7 * length(sites)))
in_train <- location %in% train_sites
train <- d[in_train, ]
test <- d[!in_train, ]
stopifnot(length(unique(train$presence)) == 2,
          length(unique(test$presence)) == 2)

table(train$presence)
table(test$presence)

write.csv(data.frame(row = seq_len(nrow(d)), in_train),
  "output/06_data_split.csv", row.names = FALSE)
```

This is a simple random split of locations. Nearby sites may still share environments, and sampling bias remains.

3. Fit both models to train only. The GLM uses standard formula for predictor terms on the logit scale. The GAM allows smooth curves; k = 4 keeps the smooths modest in this first comparison.

```
glm_fit <- glm(presence ~ bio1_c + bio12_mm, data = train, family = binomial())

gam_fit <- gam(presence ~ s(bio1_c, k = 4) + s(bio12_mm, k = 4),
  data = train, family = binomial(), method = "REML")

summary(glm_fit)
summary(gam_fit)

stopifnot(glm_fit$converged, gam_fit$converged)
```

Inspect warnings and predictor variation. Very small datasets, few distinct values, or separation can make either fit unreliable.

Part C Evaluate the same held-out rows

4. Predict scores for test data, which neither model used for fitting. Save the scores so you can trace the evaluation back to its rows.

```
test$glm_score <- predict(glm_fit, test, type = "response")

test$gam_score <- predict(gam_fit, test, type = "response")

stopifnot(all(is.finite(test$glm_score)),
          all(is.finite(test$gam_score)))

write.csv(test, "output/06_test_predictions.csv", row.names = FALSE)
```

AUC measures how often an occurrence receives a higher score than a background row. The small helper below calculates this ranking measure without another package. It requires both classes and finite scores, checked above.

```
# helper function
auc_score <- function(y, score) {
  n1 <- sum(y == 1)
  n0 <- sum(y == 0)
  ranks <- rank(score, ties.method = "average")
  (sum(ranks[y == 1]) - n1 * (n1 + 1) / 2) / (n1 * n0)
}

comparison <- data.frame(
  model = c("GLM", "GAM"),
  test_auc = c(auc_score(test$presence, test$glm_score),
               auc_score(test$presence, test$gam_score)))

comparison

write.csv(comparison, "output/06_model_comparison.csv",
  row.names = FALSE)

saveRDS(list(GLM = glm_fit, GAM = gam_fit),
  "output/06_models.rds")
```

An AUC near 0.5 indicates little ranking ability; larger values indicate stronger occurrence–background discrimination for these test data.

Pause and interpret

1. Which model has the higher test AUC? How large is the difference?
2. Why would evaluating the training data give a less convincing test?
3. Why can we not interpret background rows as confirmed absences when discussing commission error or specificity?

Part D Compare the response curves

5. Vary your first predictor over its training range and hold the second at its training median. Replace names and axis units for your project. Both curves must use exactly the same predictor values.

```
curve <- data.frame(bio1_c = seq(min(train$bio1_c), max(train$bio1_c),
              length.out = 100), bio12_mm = median(train$bio12_mm))

curve$GLM <- predict(glm_fit, curve, type = "response")
curve$GAM <- predict(gam_fit, curve, type = "response")

png("output/06_response_comparison.png", width = 1600, height = 1100, res = 170)
plot(curve$bio1_c, curve$GLM, type = "l", lwd = 2,
  ylim = c(0, 1), xlab = "Annual mean temperature (degrees C)",
  ylab = "Occurrence versus background score")
lines(curve$bio1_c, curve$GAM, col = "firebrick3", lwd = 2)
legend("topright", c("GLM", "GAM"),
  col = c("black", "firebrick3"), lwd = 2, bty = "n")
rug(train$bio1_c[train$presence == 1])
dev.off()
```

These are conditional curves; some combinations may be poorly represented in the data.

README Lab 06 prompts

1. Report the training and testing counts for occurrences and background. Explain how you kept the two subsets separate and one limitation of this split.
2. Describe one difference between the GLM and GAM response curves. Relate it to the model forms.
3. Report both test AUC values. Does the more flexible model perform better on this split? Explain what these presence–background AUC values do and do not establish.
4. For your November 6 preliminary results, which model would you carry forward, why, and what would you check next? Support your provisional choice with the curve, evaluation, and biological reasoning.

Assignment 6 submit October 30 through Canvas

Submit your private GitHub repository link to Lab 06 on Canvas by Friday October 30, 2026 at midnight. Include analysis/06_comparing_models.R, the saved split, test predictions, model comparison CSV, saved models, response comparison figure, and a README Lab 06 section. Restart R, run from the top to check for repeatability. If it works well, commit and push. Acknowledge any AI assistance; you must be able to fully understand your code and explain your decisions.